MILITARY SPECIFICATION

RIFLE, 7.62MM: M14, NATIONAL MATCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 7.62 millimeter (mm), lightweight, air-cooled, gas-operated, semiautomatic, magazine-fed, glass-bedded, military-type rifle designed to provide a refined rifle for marksmanship purposes and for use in national rifle match competitive shooting.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6868	- Inspection Process, Magnetic Particle
NIL-W-13855	- Weapons; Small Arms and Aircraft Armament Subsystems: General Specification for
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	- Inspection Equipment; Acquisition, Maintenance and Disposition of
MIL-C-45662	- Calibration System Requirements
MIL-C-46477	- Cartridge, 7.62mm, NATO, Test, High-Pressure, M60
MIL-C-46934	- Cartridge, 7.62mm, NATO, Match, M118

STANDARDS

Federal

Fed. Std. No. 356	- Commercial Packaging of Supplies and Equipment

FSC 1005

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Rock Island Arsenal

J7790476	- Rifle, 7.62MM, M14 N.M.

(Product drawings referenced in this specification form a part of Drawing J7790476.)

PUBLICATIONS

Rock Island Arsenal

P7790476	- Packaging Data Sheet for Rifle, 7.62mm, M14 N.M. w/Equipment
QE-G-DL7790476	- Quality Engineering Data List

U.S. Government

SF-32	- General Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American National Standards Institute (ANSI)

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1),

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3.2 Materials, construction and design. Rifles shall conform to the materials, construction and design requirements specified herein, on Drawing J7790476 and in MIL-W-13855.

3.2.1 Barrel and receiver assembly. The barrel shall be assembled and aligned to the receiver in the position specified on Drawing F7791363.

3.2.1.1 Barrel. The barrel shall be so fabricated that when the rifle is completely assembled, the requirements for targeting and accuracy hereinafter prescribed shall be met. The barrel shall be free of cracks and seams and the bore and chamber shall be free of foreign matter, corrosion, burrs, pits, bulges and deformations. Burrs and sharp edges shall be removed from chamber edges. Scratches or marks occurring in a chamber which otherwise meets the surface roughness requirements shall be permitted provided they do not cause marks on the case of a high-pressure test cartridge fired in the chamber. The barrel shall be proof and magnetic particle inspection marked in accordance with Drawing F7791362.

3.2.1.2 Cartridge clip guide. The cartridge clip guide shall be fastened securely to the receiver by the spring pin so that there is no relative movement. The cartridge clip guide shall position and retain the clip to allow recharging of the magazine without removing the magazine from the rifle.

3.2.1.3 Connector locking pin. The connector locking pin shall be retained in the receiver by the flared retaining pin and shall move through its full range of travel without binding (see Drawing F7791363).

3.2.1.4 Selector shaft lock, selector shaft, and sear release. The selector shaft lock, selector shaft and sear release shall be assembled and welded to the receiver in accordance with the applicable drawing. After assembly and welding, these parts shall be permanently fused to the receiver making them an integral part of the receiver and rendering the rifle incapable of automatic fire. Welded areas shall be touched up with touch-up materials in accordance with the applicable drawing and MIL-W-13855.

3.2.2 Bedding of receiver and trigger housing. The receiver and the trigger housing shall bear on the stock and stock liner contact surfaces and shall grip the stock securely so that there is no relative movement. The periphery of the rear end of the receiver shall be completely seated so that there is no overhang of the stock.

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3.2.3 Bolt assembly and operating rod action. With the firing mechanism, stock and operating rod spring disassembled from the rifle, the bolt assembly, with the operating rod connected, shall move of its own weight in either direction through its full range of travel in the receiver when the axis of the rifle is manually changed from the horizontal to approximately 60 degrees elevation and depression.

3.2.4 Bolt Assembly. The bolt assembly, with the operating rod disconnected, shall move through its full range of travel in the receiver without binding and when in the fully locked (forward) position, the bottom surface of the right hand locking lug shall seat on the receiver. Firing pin protrusion shall be .044 inch minimum to .060 inch maximum in the forward position.

3.2.4.1 Bolt and roller assembly. The bolt roller shall be retained on the bolt by the retaining ring and shall rotate without binding on the bolt shaft. The bolt and roller assembly shall be proof and magnetic particle inspection marked in accordance with Drawing J7790476.

3.2.4.2 Ejector. The ejector shall be retained in the bolt by the extractor shank and shall be held in the forward position by the ejector spring. The ejector shall move through its full range of travel under spring action without binding in the bolt and shall eject cartridge cases completely out of the rifle when the rifle is function fired or manually operated. When the rifle is function fired, the ejection pattern shall be from 15° to 90° (12:30 to 3:00 o'clock) to the right of the longitudinal axis of the rifle.

3.2.4.3 Extractor. The extractor shall be retained in the bolt by the extractor-d plunger assembly. The extractor shall move through its full range of travel under spring action without binding, and shall extract cartridge cases from the barrel when the rifle is function fired or manually operated.

3.2.4.4 Firing pin. The firing pin shall be retained in the bolt by the extractor shank and shall move of its own weight through its full range of travel in the bolt.

3.2.5 Bolt stop. The bolt stop shall be retained on the receiver by the spring pin and shall be held in the "down" position by action of the bolt stop spring. The bolt stop shall move through its full range of travel, and when positioned manually or by action of the magazine follower, it shall hold the bolt in the open (rearward) position. The bolt stop shall disengage from the bolt by action of the bolt stop spring when the magazine follower is disengaged, and the operating rod is retracted.

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3.2.6 Connector assembly. The connector assembly shall be retained on the receiver by the connector locking pins and spring action of the connector plunger. The connector assembly shall prevent the operating rod handle from popping out of the receiver way.

3.2.7 Flash suppressor. The flash suppressor shall be aligned as specified on the applicable drawing and shall be fastened securely to the barrel by the flash-suppressor"nut which in turn shall be fastened securely by a set screw. There shall be no relative movement of the flash suppressor, the flash suppressor nut or the set screw.

3.2.8 Firing mechanism. The firing mechanism shall assemble to the rifle without the use of tools, and shall disassemble from the rifle when the trigger guard is unlatched from the trigger housing using a tool simulating a cartridge.

3.2.8.1 Hammer. The hammer shall pivot on the hammer pin through its full range of travel without binding, and when the hammer is released from the cocked position it shall pivot forward to the stop position on the trigger housing under spring action.

3.2.8.2 Safety. The safety shall be movable manually from one position to the other, and shall remain in the position set until reset manually. The safety shall block the trigger and lock the hammer in a position out of contact with the trigger when the hammer is cocked and the safety is moved to the "Safe" (rearward) position. When the safety is moved to the "Fire" (forward) position, the hammer shall release when the trigger is pulled.

3.2.8.3 Sear. When the trigger is retained rearward, and the hammer is retracted, the sear shall engage and hold the hammer until the trigger is released and allowed to move forward.

3.2.8.4 Trigger. The trigger hooks shall hold the hammer in the cocked position until the trigger is pulled, and after partial or completed trigger pull the trigger shall return to its normal forward position under spring action.

3.2.8.5 Trigger guard. The trigger guard shall latch to and unlatch from the housing, and when unlatched and rotated shall be capable of cocking the hammer. The latches shall securely retain the firing mechanism to the rifle.

3.2.9 Gas cylinder group. The gas cylinder shall be retained on the barrel by the gas cylinder lock and plug. There shall be no rotational or longitudinal movement of these parts. The gas piston shall

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move of its own weight through its full range of travel in the gas cylinder with the rifle axis vertical. The gas cylinder plug shall be assembled with the torque specified on Drawing J7790476. The gas cut-off valve shall rotate from one position to the other only when manually pressed inward and rotated, and when indexed and released shall return to the outward position by spring action.

3.2.10 Handguard assembly. The handguard assembly shall be retained on the barrel by the rear handguard band and the front band.

3.2.11 Magazine assembly and magazine latch. The magazine follower shall function through Its full range of travel under spring action without binding. When the last cartridge from the magazine has been fired, the follower shall position the bolt stop to hold the bolt open. The magazine shall assemble to the rifle without manual depression of the magazine latch and shall be retained in the magazine-well of the receiver by the magazine latch and the operating rod spring guide. When the magazine latch is pressed forward, it shall permit manual removal of the magazine from the rifle,

3.2.12 Operating rod. The operating rod shall function through its full range of travel without binding. when manually operated, the operating rod shall not contact the stock in the area where the handle passes through the opening between the stock and the barrel, and shall not have contact with the front corner of the receiver throughout the full range of travel of the operating rod. When in the full retracted position, the operating rod shall bear lightly against the underside of the enlarged section of the barrel.

3.2.12.1 Operating rod, two piece design. The operating rod tube shall be securely welded to the handle in accordance with Drawing D7267064.

3.2.13 Operating rod guide. The operating rod guide shall be retained on the barrel by the spring pin, and there shall be no relative movement of the operating rod guide.

3.2.14 Operating rod spring and spring guide. With the magazine assembled to the receiver, the operating rod spring shall seat on the operating rod spring guide, and shall not seat on the receiver.

3.2.15 Sights.

3.2.15.1 Front sight. The front sight shall be capable of lateral adjustment on the tenon of the flash suppressor, and shall be fastened securely by the front sight screw. There shall be no relative movement of these parts.

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3.2.15.2 Rear sight assembly. The windage knob and the elevating knob shall function without binding through their full range of travel, and shall produce positive retention perceptible by touch for each detent notch when the sight is adjusted manually. The aperture shall function in the base without binding, and the rear sight cover shall bear on the aperture when the elevation knob is rotated manually. The aperture, when raised to the 600-meter position shall not have any perceptible looseness when pressure is applied in a horizontal plane along the windage axis. When the aperture is in a vertical plane away from the rifle, the spring tension of the cover shall return the aperture to its original position. The eyepiece shall rotate manually on the aperture 360 degrees without binding, and shall produce positive retention perceptible by touch for each 180 degrees of rotation. The windage knob shall fit in the base without perceptible backlash.

3.2.16 Stock assembly.

3.2.16.1 Plate assembly with shoulder rest. The plate assembly with shoulder rest shall be fastened securely to the stock by the butt screws with no overhanging metal. The butt plate cap shall be retained in the open position by the action of the butt plate plunger and spring, and shall be retained in the closed and locked position by the plunger operated catch. The shoulder rest plate shall be retained in the open or closed position by the action of the ball and the lock spring. Each stock shall have access through the butt plate to the accessory holes. The holes shall accept the go-gage depicted on Drawing B7271641.

3.2.16.2 Stocks. The grain in the wood stocks shall be straight, and in the longitudinal direction of the component. Light streaks in the wood shall be treated to blend with the remainder of the component to produce a uniform color. Patching of stocks shall not be allowed, except that filling of small pin holes not exceeding 0.06 inch in diameter will be allowed in the receiver well area from the centerline of the stock liner screw holes back toward the butt end 5.0 ± 1.0 inches, and forward from the centerline of the stock liner screw holes 6.0 ± 1.0 inches.

3.2.16.3 Stock ferrule and stock liner assembly. The stock ferrule shall be securely assembled to the stock in accordance with the applicable drawing, and the stock liner assembly shall be securely fastened to the stock by the screws. There shall be no relative movement of these parts.

3.2.16.4 Receiver and stock clearance. The clearance between the receiver rails and stock bedding surface shall be as depicted on Drawing F1101O262, Section C-C.

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3.2.16.5 Front band and stock ferrule clearance. There shall be 1/64-inch minimum clearance between the front band and stock ferrule (see Drawing J7790476).

3.2.16.6 Swivels. The butt swivel shall be fastened securely to the stock by the long butt plate screw. The sling swivel assembly shall be fastened securely to the stock by rivets. There shall be no relative movement of these parts except that the swivel loop shall rotate using light finger pressure.

3.2.17 Soundness of weld. The welding of the selector shaft and the sear release to the receiver shall show good fusion. The weld and welded parts shall be free of cracks, porosity, inclusions and other metallic discontinuities, Testing shall be as specified in 4.5.10.2.

3.2.18 Tightness of draw. When the barrel is drawn into the receiver, the breakaway torque shall not be less than 80-foot-pounds. Testing shall be as specified in 4.5.10.3.

3.2.19 Headspace. After proof firing, the headspace shall be 1.6355 inches minimum to 1.6385 inches maximum. Testing shall be as specified in 4.5.10.4.

3.3 Performance characteristics.

3.3..1 Firing pin indent. Firing pin indent shall be .016 inch minimum, shall not cause pierced primers, and shall not be off center more than one-half the diameter of the firing pin point. Testing shall be as specified in 4.5.10.5.

3.3.2 Trigger pull. The trigger pull shall be free of creep, and shall be within the range of 4 1/2 to 6-pounds. Creep shall be interpreted to mean any perceptible rough movement between the time the trigger slack is taken up and the hammer is released. Testing shall be as specified in 4.5.10.5.

3.3.3 High-pressure resistance. Rifles shall withstand the proof firing of one Government standard 7.62mm. M60 hiqh-pressure test cartridge. Cartridges shall be in accordance with MIL-C-46477. Testing shall be as specified in 4.5.10.6.

3.3.4 Functioning. Rifles shall be capable of only semiautomatic fire, and shall operate without malfunctions or unserviceable parts. Rifles shall be function fired using Government standard 7.62mm, M118, NATO match cartridges. Cartridges shall be in accordance with MIL-C-46934. Testing shall be as specified in 4.5.10.6.

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3.3.5 Targeting and accuracy. Ammunition used for targeting and accuracy firing shall be Government standard 7.62mm, M118 NATO match cartridges in accordance with MIL-C-46934.

3.3.5.1 100 yards. The average group size of three individual ten-shot targets fired at 100 yards shall not exceed 3.5 inches. The extreme spread of any one individual ten-shot group shall not be greater than 5.0 inches, and each ten-shot group shall be within the targeting area specified on Drawing C7790136. Testing shall be as specified in 4.5.10.6.

3.3.5.2 100 meters. The average group size of three individual ten-shot targets fired at 100 meters shall not exceed 3.8 inches. The extreme spread of any one individual ten-shot group shall not be greater than 5.5 inches, and each ten-shot group shall be within the targeting area specified on Drawing C7790136. Testing shall be as specified in 4.5.10.6.

3.4 Marking. Rifles shall be marked in accordance with MIL-W-13855. Rifles shall be identified by individual serial numbers assigned by the procuring activity (see 6.1).

3.5 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the-inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall perform, as a minimum, inspections in accordance with the specifications, quality assurance provisions and the contract. These minimum inspections shall not be construed as relieving the contractor of his responsibilities under terms of the contract to furnish the Government with items complying with and conforming to the requirements of the contract, drawings and specifications.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.2.1 Commercial items, for purposes herein, are considered to be those items which are industry developed and manufactured and are available off-the-shelf to industry, the Government, and/or the general public.

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4.3 Classification of inspection. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified (see 6.1), the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. When" Acceptable Quality Levels (AQL's) are" specified, the AQL's shall be individually applied in accordance with MIL-STD-105.

4.5.2 In-Process control. The contractor shall establish in-process inspection at strategically located points throughout his manufacturing processes to assure continuous control of product quality. Except for unmodified Military standard and commercial parts (see 4.2.1 and 4.5.4.2), the contractor's inspection system shall provide for inspection and approval of the first piece at each operation and/or the finished part before quality production. In addition, the contractor shall provide and maintain work gages and other measuring and testing devices necessary to accomplish inspection and control quality during his manufacturing processes.

4.5.3 Inspection records. Contractor's records of inspection shall be accurate, complete and available to the Government upon request. The contractor's records of inspection shall provide, as a minimum, the following information:

a. Contract order number
b. Drawing and specification number with revision letter and revision date
c. Nomenclature of items inspected
d. Number of pieces inspected
e. Individual characteristics inspected
f. Conformance/nonconformance criteria
g. Description and quantity of defects found

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h. Results of inspection (e.g. identification of conforming and of nonconforming product). Actual inspection and test results shall be recorded when actual measurements and tests were the basis for conformance/nonconformance decisions.
i. Date of inspection
j. The following shall be included when sampling is used:

(1) Sampling plan
(2) AQL's used
(3) Lot size and lot number
(4) Sample size

4.5.3.1 The contractor shall make available to the Government representative records of qualification of operators and equipment for special processes (i.e. magnetic particle, radiographic, etc.) to applicable specifications and standards. This shall apply whether the special processes are performed at the subcontractor's facility or at the contractor's plant, All inspection records, including certified test reports, operator/process certifications, test reports, reports of visits to the subcontractor, etc., shall be maintained on file at the contractor's plant and made readily available to the Government representative, upon request, for the duration of the contract and for three years thereafter.

4.5.4 Certification provisions.

4.5.4.1 Certified test reports (CTR). When specified in the contract or in documents referenced therein, the contractor shall make available to the Government a certified test report for each lot of parts, assemblies, subsystems and systems by lot number prior to acceptance. Certified test reports are NOT required for Military standard and commercial items (see 4.2.1 and 4.5.4.2). This test report is in addition to, and not in lieu of, any rights of the Government under this contract or law. A CTR may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of the contract item(s). As a minimum, the report shall contain the following:

a. Name of company and date.
b. Contract number or purchase order number, national stock number and drawing number.
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment shall be given.
d. All inspections and tests required by contract (i.e., material, processes, performance, functional, etc.) shall be recorded in test reports. These reports shall identify each lot, submitted for acceptance by lot number, the specification or drawing, revision and date, grade or type as applicable, number of specimens tested, specified characteristics and requirements, and actual results obtained.

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e. Reports of the raw material producer's chemical, mechanical and physical analyses,
f. A statement, as follows, certifying that material meets all requirements of the contract.

"The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: All of the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not limited to the drawings and specifications); the inspection and test results, and the analyses appearing herein are true and accurate; and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for payment."

g. Signature and title of certifying official.

4.5.4.2 Certificate of conformance (COC). A Certificate of Conformance (COC) is required for Military standard and commercial items (see 4.2.1 and 4.5.4.1). When a Certified Test Report (CTR) is NOT required for material and process specification requirements, a COC, supported by inspection and test data, material analyses or certification from the raw material producer or processor, shall be made available to the Government for specifications covering raw material and processes. The contractor shall make the COC available to the Government prior to or with the request to perform acceptance inspection approval by the Government. This is in addition to, and not in lieu of, any rights of the Government under this contract or law. A COC may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of contract item(s). As a minimum, the COC shall contain the following:

a. Name of company and date.
b. Contract number or purchase order number, national stock number, and drawing number,
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment shall be given.
d. A statement, as follows, certifying that material meets all requirements of the contract:

"The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: All of the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not 1 imited to the drawings and specifications); the analyses appearing herein are true

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and accurate analyses; and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for such payment."

e. Signature and title of certifying official.

4.5.5 Application of quality assurance provisions. The inspection provisions specified should be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspection of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. In any event, the Government reserves all rights under General Inspection Article 5, Standard Form 32.

4.5.6 Quality engineering inspection methods. The following methods apply as applicable:

a. Where "Visual" is specified as the inspection method for protective coating, the coating shall be visually examined for completeness, uniformity in appearance and color, and for freedom from pits, corrosion, scratches, and worn or bare spots.
b. Where "Visual" is specified as the inspection method for dimensional inspection, the characteristic shall be either scaled, or compared with a specimen of known acceptable quality that has been established as an inspection standard, for conformance to the requirements of the drawing.
c. Where "Visual" is specified as the inspection method for assemblies, the assembly shall be visually examined for completeness, security, function and conformance to specified requirements. When possible, the functioning of the assembly shall be inspected by manual operation.
d. Where "Visual" is specified as the inspection method for surface roughness values, a comparison standard conforming to ANSI B46.1 shall be used. If determination of surface roughness by comparison standard is questionable, a surface measuring instrument shall be used.
e. Where "SMTE" (Standard Measuring and Test Equipment) is specified as the method of inspection, the contractor may use any type industry-developed, commercially available, multi-usage equipment or special inspection and/or testing equipment approved by the Government, (Previously identified as; "SME", Standard Measuring Equipment and "STE". Standard Test Equipment.)

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f. When Government Furnished Acceptance Inspection and Test Equipment is specified as the method of inspection in the contract or on the Data Lists, the contractor shall use inspection equipment fabricated in accordance with the Government drawing referenced, This requirement takes precedence over the Inspection Method in Table (see h. below).

g. Where "STM" (Special Test Method) is specified as the method of inspection, the methods and procedures shall be as specified.

h. Alternative inspection methods and inspection equipment may be used by the contractor when such methods and equipment equal or exceed the specified accuracy and provide, as a minimum, the quality assurance required in the contractual documents. Prior to applying such alternative inspection methods and inspection equipment, the contractor shall describe them in a written proposal and shall demonstrate, for the approval of the Procuring Contracting Officer (PCO), that their effectiveness is equal to, or better than, the contractual quality assurance method or equipment.

4.5.7 Acceptance inspection and test equipment. INSPECTION AND TEST EQUIPMENT USED SHALL BE CAPABLE OF REPEATABLE MEASUREMENTS, BY VARIOUS EXPERIENCED INSPECTION/TEST PERSONNEL, TO AN ACCURACY OF 10% OF THE TOTAL TOLERANCE OF THE CHARACTERISTIC BEING INSPECTED.

In the event the contractor desires relief from this requirement for electrical testing, a technically supported request for relief or waiver shall be submitted thru the QAR and the Defense Contract Administration Services Region (DCASR) to the PCO.

a. The contractor is responsible for providing all standard measuring and test equipment required.

b. Required Government designed inspection and test equipment is shown on Quality Engineering Data List QE-G-DL7790476 and all documents listed thereon.

c. The contractor is responsible for requisitioning the Government designed inspection and test equipment shown as available in the contract.

d. The Government designed inspection and test equipment shown as not available in the contract shall be provided by the contractor in accordance with MIL-I-45607. This equipment shall be in accordance with the Government drawings.

e. The contractor is responsible for providing such quantities of all inspection and test equipment, calibrated in accordance with MIL-C-45662, as required to support:

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(1) Similar multiple inspection at the contractor's facilities
(2) Inspection operations at sub-contractor's facilities
(3) Required periodic calibration, maintenance or replacement.

4.5.7.1 Equipment utilization. Use of contractor's or Government furnished inspection equipment, when desired by the Government representative, shall be permitted without charge. Government-furnished acceptance inspection equipment shall not be used by the contractor in lieu of contractor work gages and equipment.

4.5.7.2 Inspection standards. Inspection standards shall be utilized for those characteristics requiring inspection decisions by visual (eye sight) means. Items selected as visual comparison standards shall be mutually agreed to by the contractor and the Government, within drawing and specification requirements, and shall be used to assist in determining configuration and minimum acceptance criteria. The visual comparison standards selected shall be subject to approval by the responsible technical agency. Each comparison or inspection standard shall be kept under the control of the contractor's inspection element and be positively identified as to the characteristic or condition the standard represents, date established as the standard, number of the standard, and identity of the contractor and the Government inspection personnel establishing the standard,

4.5.7.3 Ammunition. Unless otherwise specified in the procurement documents (see 6.1) Government standard 7.62mm, M60, high-pressure test cartridges (see MIL-C-46477) shall be used for the high-pressure resistance test (see 4.6.6) and Government standard 7.62mm, M118, NATO match cartridges (see MIL-C-46934) shall be used for the functioning and targeting and accuracy firing test (see 4.6.7).

4.5.8 Parts inspection. Inspection of parts and subsequent assemblies shall be as specified in the contract (see 6.1), and shall be accomplished prior to assembly into the next higher assembly or the end item.

4.5.9 Examination. Each rifle shall be examined as specified in Table 1. Rifles failing to meet the requirements shall be rejected.

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TABLE I

EXAMINATIONS

Item	Requirement	Inspection Method
Materials, construction and design	3.2	Visual
Barrel	3.2.1.1	Visual, SMTE
Cartridge clip guide	3.2.1.2	Visual
Connector locking pin	3.2.1.3	Visual
Selector shaft lock, selector shaft and sear release	3.2.1.4	Visual
Bedding of receiver and trigger housing	3.2.2	Visual
Bolt assembly and operating rod action	3.2.3	Visual
Ejector	3.2.4.2	Visual
Extractor	3.2.4.3	Visual
Bolt stop	3.2.5	Visual
Firing mechanism	3.2.8	Visual
Hammer	3.2.8.1	Visual
Safety	3.2.8.2	Visual
Sear	3.2.8.3	Visual
Trigger	3.2.8.4	Visual
Trigger guard	3.2.8.5	Visual
Handguard assembly	3.2.10	Visual
Magazine assembly and magazine latch	3.2.11	Visual
Operating rod	3.2.12	Visual
Operating rod, two piece design	3.2.12.1	Visual
Operating rod guide	3.2.13	Visual
Plate assembly with shoulder rest	3.2.16.1	Visual
Stock ferrule and stock liner assembly	3.2.16.3	Visual
Swivels	3.2.16.6	Visual
Marking	3.4	Visual
Workmanship	3.5	Visual

4.5.9.1 Barrel and receiver assembly. Each barrel and receiver assembly shall be examined for allgnment of the barrel to the receiver requirement as specfied on Drawing F7791363 (see 3.2.1).

4.5.9.2 Rifle disassembly. Each rifle shall be disassembled and examined as follows:

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4.5.9.2.1 Stock assembly. Visually examine the stock to assure freedom from mutilations and wood defects (see 3.2.16.2). Examine access to the accessory storage holes in the stock, with the butt plate assembled, using inspection equipment in accordance with Drawing B7271641 (see 3.2.16.1).

4.5.9.2.2 Operating rod spring and spring guide. Visually examine the spring to assure freedom from damage and cracked or broken ends, and the spring guide for damage and burs. Examine for the proper seating of the operating rod spring on the spring guide (see 3.2.14).

4.5.9.2.3 Connector assembly. Visually and manually examine the connector assembly for proper operation (see 3.2.6). Examine the connector assembly for damaged parts. Disassemble the operating rod from the receiver and examine for proper assembly, broken parts, and surface defects in the bolt roller cam surfaces and the receiver way lugs.

4.5.9.2.4 Bolt assembly. Visually and manually examine the bolt assembly for proper operation (see 3.2.4). Disassemble the bolt assembly from the receiver and visually examine for damage and surface defects and deformations on the bolt lugs. Examine firing pin protrusion and intrusion using inspection equipment in accordance with Drawing D7796327 (see Drawing C7790187).

4.5.9.2.5 Firing Visually arid manually examine the firing pin for proper operation see 3.2.4.4), damage, surface defects, scoring caused by burs in the bolt, and mutilations on the point. Visually examine the ejector and extractor lip for damage and surface defects.

4.5.9.2.6 Bolt and roller assembly. Visually and manually examine the bolt and roller assembly for proper operation (see 3.2.4.1), damage and surface defects. Visually examine for proof and magnetic particle inspection marking (see 3.4 and Drawing J7790476).

4.5.9.2.7 Barrel and receiver assembly. Visually examine the receiver assembly for proof and magnetic particle inspection marks, “N.M.” marking on receiver,” and that windage graduations are clear and legible (see 3.2.1.1, 3.4 and Drawings F7790189 and J7790476).

4.5.9.2.8 Front sight. Manually inspect front sight for lateral adjustment and security of fit (see 3.2.15.1). Visually examine to assure there is no overhang on the tenon of the flash suppressor. Visually examine the sight wings for proper profile and damage or improper surface finish.

4.5.9.2.9 Rear sight assembly. Visually and manually examine the rear sight for proper operation, proper profile, backlash, any damage, proper surface finish and excess oil (see 3.2.15.2).

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4.5.9.2.10 Gas cylinder group. Manually examine the gas cylinder group for proper operation (see 3.2.9). Visually examine the gas cylinder and plug threads for damage.

4.5.9.2.11 Flash suppressor. Manually examine secure retention of the flash suppressor to the barrel (see 3.2.7). Visually examine for damage and proper surface finish.

4.5.9.3 Rifle assembly. Examine reassembly of each rifle to assure that the rifle can be assembled without use of hand tools (except for assembly of the extractor to the bolt).

4.5.9.3.1 Flash suppressor alignment. Examine alignment of the flash suppressor to the barrel using inspection equipment in accordance with Drawing C11015429 and C1101543O (see 3.2.7).

4.5.9.3.2 Gas cylinder group. Examine alignment of the gas port holes in the gas cylinder, the gas cut-off valve, and the barrel usinq inspection equipment in accordance with Drawing A11015316, Examine tie gas cylinder lock to verify assembly requirements of Drawing J7790476 (see 3.2.9). Examine reassembly of the gas cylinder plug using the assembly device in accordance with Drawing C11015427 and verify the torque limits specified on Drawing J7790476.

4.5.9.3.3 Magazine and receiver concentricity. With the magazine assembled to the rifle, examine concentricity of the cartridge clip guide with the magazine opening in the receiver using inspection equipment in accordance with Drawing D11015522 (see 3.2.11).

4.5.9.3.4 Receiver and stock clearance. Visually examine clearance between the receiver rails and stock bedding surface (see area forward of the stock horseshoe, Section C-C of Drawing F11010262, Dimension .00 + .01 and 3.2.16.4).

4.5.9.3.5 Front band and stock ferrule clearance. Visually examine 1/64 inch minimum clearance between the front band and the stock ferrule (see Drawing J7790476 and 3.2.16.5).

4.5.9.3.6 Ammunition loading, extraction and ejection. Insert a dummy round in the magazine before inserting the maqazine to assure that It will assemble to the rifle without manual depression of the magazine latch (see 3.2.11). Manually draw the operating rod and bolt to the open (rearward) position and allow to return to the closed position under spring action to examine proper stripping of the cartridge from the magazine by the bolt. Manually withdraw the operating rod and bolt to the open position to determine proper extraction and ejection of the cartridge (see 3.2.4.2 and 3.2.4.3).

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4.5.9.3.7 Semiautomatic fire. With the magazine removed, charge the rifle (pull the bolt) while depressing the trigger, then push forward on the rear of the connector and verify that the rifle will not fire automatically (see 3.2.1.4).

4.5.10 Testing.

4.5.10.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete rifle. If test requirements cited herein are not met, acceptance of the rifle shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical, and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine rifles, partially assembled rifles, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned rifle for retest.

4.5.10.2 Soundness of weld testing. Each receiver or barrel and receiver assembly shall be tested for soundness of weld (see 3.2.17) using the test method specified in 4.6.1. Welds or parts failing to meet the requirements shall be rejected.

4.5.10.3 Tightness of draw testing. A sample of three barrel and receiver assemblies shall be selected from each days production and tested for tightness of draw (see 3.2.18) using the test method specified in 4.6.2. Failure of any assembly in the sample to meet the requirements shall cause rejection of the represented lot. Test frequency may be reduced to not less than a test of three assemblies from each week's production when five consecutive days production have been acceptable. If a failure occurs during reduced inspection, revert back to the original inspection plan.

4.5.10.4 Heads ace testing. Each rifle shall be tested for headspace (see 3.2.19 using the test method specified in 4.6.3. Rifles failing to meet the requirements shall be rejected.

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4.5.10.5 Firing pin indent and trigger pull testing. A sample of ten rifles shall be selected from each lot and tested for firing pin indent (see 3.3.1) and trigger pull (see 3.3.2) using the test methods specified in 4.6.4 and 4.6.5 respectively. Failure of any rifle in the sample to meet the requirements shall cause rejection of the represented lot.

4.5.10.6 High-pressure resistance, functioning, and targeting and accuracy firing testing. Each rifle shall be tested for high-pressure resistance (see 3.3.3), functioning (see 3.3.4), and targeting and accuracy (see 3.3.5) using the test methods specified in 4.6.6 and 4.6.7 respectively. Rifles failing to meet the requirements shall be rejected.

4.5.11 Inspection for acceptance. The contractor shall perform inspection for acceptance upon completion of all manufacturing and in-process inspection operations, prior to preservation and packing. In-spection for acceptance shall include, but is not limited to, inspection for workmanship, protective finish, missing features (missing operations), characteristics not previously inspected, and characteristics that may have been altered or changed as a result of subsequent operations after prior inspections were completed.

4.5.12 Inspection of packaging. Unless otherwise specified (see 6.1), inspection to determine compliance with preservation, packing and marking requirements of the applicable packaging documentation, for the level designated in the contract, shall be as specified in MIL-P-14232.

4.6 Test methods..

4.6.1 Soundness of weld test. Magnetic particle inspection for soundness of weld (see 4.5.10.2) shall be Performed in accordance with MIL-I-6868. Magnetization shall be by the-continuous process using the circular and longitudinal methods. The longitudinal method shall be accomplished using a standard five turn magnetizing coil. Inspection shall be by the wet process using a fluorescent suspension. When magnetic particle inspection is to be performed on barrel and receiver assemblies, the current magnitude shall be 800 to 1200 amperes. When magnetic particle inspection is to be performed on receivers, the current magnitude shall be 400 to 500 amperes.

4.6.2 Tightness of draw test. The barrel and receiver assemblies shall be tested for tightness of draw (see 4.5.1O.3) using inspection equipment in accordance with Drawing F7271792. The barrel and receiver holding fixtures shall be secured to a bench or other suitable surface. The barrel and receiver assembly shall then be secured in the fixture as specified on the applicable drawing and torque applied to the receiver in the direction to disassemble it from the barrel.

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4.6.3 Headspace test. Rifles shall be gaged for headspace (see 4.5.10.4) after proof firing, prior to acceptance, using the inspection equipment in accordance with Drawing F7274733. The rifle shall be held in a horizontal position (approximately) with the bolt open. The spring loaded plug shall then be inserted in the barrel chamber and the bolt manually returned to battery position. The indicator portion of the gage shall be set to zero using the setting check as specified on the applicable drawing, and then inserted in the barrel from the muzzle end until the bodies of the indicator gage and the spring loaded plug come in contact. The bolt shall be fully locked while taking the indicator gage reading and the reading shall determine whether the headspace is within the requirements.

4.6.3.1 In the event of a dispute over the test results when tested as specified in 4.6.3, the referee test method shall be as specified in 4.6.3 except that the rifle shall be gaged with the operating rod disconnected from the bolt.

4.6.4 Firing pin indent test. Rifles shall be tested for firing pin indent (see 4.5.10.5) using inspection equipment in accordance with Drawings D7271741 and C7318984. The rifle shall be held in a horizontal position (approximately) with the bolt held open and the copper compression cylinder shall be inserted in the barrel chamber. The bolt shall be manually returned to battery position and the trigger pulled to release the hammer and indent the copper cylinder. The holding fixture shall be removed from the rifle and the depth of the indent in the copper cylinder computed by measuring the distance from the original surface of the copper cylinder (before indentation) to the bottom of the firing pin impression.

4.6.5 Trigger pull test. Rifles shall be tested for trigger pull (see 4.5.10.5), using inspection equipment in accordance with Drawings D11015517 and D11015518. The rifle shall be cocked and the safety shall be in the “fire” position when each load is gradually applied to the trigger in the direction specified on the applicable drawing. When the minimum load is applied, the hammer shall not release and when the maximum load is applied the hammer shall release. The trigger pull shall also be tested for creep by applying pressure manually to the trigger at a uniform rate of increase over a period of not less than 3 seconds.

4.6.6 High-pressure resistance test. Rifles shall be tested for high-pressure resistance (see 4.5.10.6) by firing one Government standard high-pressure test cartridge (see 4.5.7.3) in each rifle. Proof firing shall be accomplished with the rifle held in a firing stand simulating shoulder firing and conforming to Drawing F7273901. Each receiver shall have been magnetic particle inspected, and each barrel shall have been proof fired and magnetic particle inspected, in accordance with the applicable drawings, before assembly to the rifle for this test, After proof firing the rifles shall be visually examined for cracks, deformations

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and other evidence of damage, and cartridge cases shall be visually examined for bulges, splits, rings and other defects caused by defective barrels. The receiver and the bolt and roller assembly of each rifle shall be magnetic particle inspected in accordance with the applicable drawings after completion of all firing tests (high-pressure resistance, functioning and targeting and accuracy). Proof marks and magnetic particle inspection marks shall be applied, as indicated on the applicable drawings, on rifles that have passed this test.

4.6.7 Functioning and targeting and accuracy firing test. Testing of rifles for functioning and targeting and accuracy at a range of 100 yards or at a range of 100 meters (see 4.5.10.6) shall be accomplished with the rifles held in a targeting jack simulating shoulder firing and conforming to Drawing F6511841. The ammunition shall be in accordance with 4.5.7.3. Before rifles are fired for targeting and accuracy, ten shots shall be fired off target to seat the rifle and settle the rifle action. The rear sight shall be set at zero windage, the aperture elevated eight clicks from the lowest position, and the rifle sights aligned at 6 o'clock on the sighting image specified on the targeting and accuracy diagram (Drawing C7790136). Three series of ten shots shall be fired at three separate targets. During the ten rounds fired to seat the rifle and the 30 rounds fired for targeting and accuracy, rifles shall be checked to assure that there are no malfunctions or unserviceable parts and that rifles are capable of only semiautomatic fire. The target shall then be checked to determine whether the targeting and accuracy requirements have been met. Extreme spread shall be measured to the nearest one-tenth of an inch using the center of the bullet holes as reference points. The front sight shall not be filed or bent but may be moved as necessary provided that it does not overhang the tenon on the flash suppressor. After satisfactory completion of the targeting and accuracy firing test the front sight shall be locked in place by the front sight screw, and the rear sight elevating knob shall be set at the 100-meter graduation mark when the aperture is elevated eight clicks from the lowest position.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of a complete rifle preserved in accordance with Packaging Data Sheet P7790476 for the level of protection specified in the contract (see 6.1), packed Level B and forwarded as specified in 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P7790476 for the level of protection specified in the contract (see 6.1).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

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5.4 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other packaging requirements, as specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Lists of drawings, specifications and publications pertinent to the rifle, showing applicable revision dates.
c. Shipping instructions for first article (see 3.1, 4.4 and 5.1)
d. Assigned block of serial numbers for quantity of rifles on order (see 3.4)
e. Examination and testing criteria for component parts and concurrent repair parts (see 4.5.8)
f. Packaging examination and testing, if different (see 4.5.12)
g. List of acceptance inspection equipment to be furnished to the contractor (see 4.5.7) and responsibilities for other Government property to be furnished to the contractor.
h. Responsibilities for furnishing ammunition (see 4.5.7.3)
i. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3)
j. Packaging instructions for repair parts (see 5.4)

6.2 Supersession data. This specification includes the requirements of Springfield Armory Purchase Description SAPD-260 dated 6 December 1963.

Custodian:
Army - WC
Navy - OS

Preparing activity:
Army - WC

Project number:
1OO5-A51O

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-45979; Rifle, 7.62MM: M14, National Match

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes" in what way?)*

4. REMARKS

5. SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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